Do Now · retrieval

Modularity, functions & procedures

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Which loop is most suitable when the number of repetitions is known in advance?

Q2

In C#, what does && mean, and when is the whole expression true?

Q3

What is the largest unsigned 8-bit denary value?

Q4

A program reads a mark, repeats through five scores and uses if to count passes. Name the three programming constructs present.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Modularity breaks a problem into parts — A large program is easier to understand when separate jobs are placed in separate subroutines.

Core idea 2

A procedure performs a job — A C# procedure-style method uses void because it does not return a value to the caller. — CODE:

```
public static void Greeting(string name)
{
Console.WriteLine("Hello " + name);
}
```

Core idea 3

A function returns a value — The return type replaces void . A return statement sends the result back to the calling code. — CODE:

```
public static int Triple(int number)
{
return number * 3;
}
```

Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Core idea 4

Parameters receive data — A parameter is the named input in the method definition. An argument is the actual value supplied when the method is called. — CODE:
public static int Triple(int number)
↑ parameter

Triple(7)
↑ argument

Vocabulary · use the correct technical terms

Modularity — Breaking a program into smaller, separate components.

Subroutine — A named block of code that can be called to perform a task.

Method — The C# term used for the functions and procedures in this lesson.

Function — A subroutine that returns a value.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Build an IsEven function

Follow the worked method

1. Choose the return type — CODE:
public static bool IsEven(int number)
{
} — The result is true/false, so the return type is bool .

2. Use it from Main — CODE:
public static void Main()
{
bool result = IsEven(12);
Console.WriteLine(result);
} — The argument 12 is received by the parameter number .

3. Use contrasting values — CODE:
IsEven(12) → True
IsEven(7) → False
IsEven(0) → True

Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Function or procedure?

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

DisplayTitle procedure
Create a procedure-style method called DisplayTitle() .
It should output OCR H446 Computer Science .
Call it from Main() twice.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 2

Refactor a program into modules
Write a program that asks for two integers and then:
1. calls Add(int first, int second) and outputs the returned total;
2. calls ShowLine() before and after the result.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A function returns a value to its caller.

Use key vocabulary from memory

1. Explain Modularity in your own words.

2. Explain Subroutine in your own words.

3. Explain Method in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.